



Clustered Colouring of Odd- H -Minor-Free Graphs

Robert Hickingbotham, Dong Yeap Kang, Sang-il Oum, Raphael Steiner,
David R. Wood

Abstract The clustered chromatic number of a graph class $\mathcal{G}$ is the minimum integer c such that every graph $G \in \mathcal{G}$ has a c -colouring where each monochromatic component in G has bounded size. We study the clustered chromatic number of graph classes $\mathcal{G}_H^{\text{odd}}$ defined by excluding a graph H as an odd-minor. How does the structure of H relate to the clustered chromatic number of $\mathcal{G}_H^{\text{odd}}$? We adapt a proof method of Norin, Scott, Seymour and Wood (2019) to show that the clustered chromatic number of $\mathcal{G}_H^{\text{odd}}$ is tied to the tree-depth of H .

Robert Hickingbotham

School of Mathematics, Monash University, Melbourne, Australia

e-mail: rd.hickingbotham@gmail.com

Research supported by an Australian Government Research Training Program Scholarship.

Dong Yeap Kang

Extremal Combinatorics and Probability Group, Institute for Basic Science, Daejeon, South Korea

e-mail: dykang.math@ibs.re.kr

Research supported by the Institute for Basic Science (IBS-R029-Y6).

Sang-il Oum

Discrete Mathematics Group, Institute for Basic Science, Daejeon, South Korea

e-mail: sangil@ibs.re.kr

Research supported by the Institute for Basic Science (IBS-R029-C1).

Raphael Steiner

Institut für Theoretische Informatik, ETH Zürich, Switzerland

e-mail: raphaelmario.steiner@inf.ethz.ch

Research supported by an ETH Zurich Postdoctoral Fellowship.

David R. Wood

School of Mathematics, Monash University, Melbourne, Australia

e-mail: david.wood@monash.edu

Research supported by the Australian Research Council.

1 Introduction

We consider simple, finite, undirected graphs G with vertex-set $V(G)$ and edge-set $E(G)$. See [3] for graph-theoretic definitions not given here. Let $\mathbb{N} := \{1, 2, \dots\}$ and $\mathbb{N}_0 := \{0, 1, \dots\}$. For $n \in \mathbb{N}$, let $[n] := \{1, \dots, n\}$.

A *colouring* of a graph G is a function that assigns one colour to each vertex of G . For an integer $k \geq 1$, a *k -colouring* is a colouring using at most k colours. A colouring of a graph is *proper* if each pair of adjacent vertices receives distinct colours. The *chromatic number* $\chi(G)$ of a graph G is the minimum integer k such that G has a proper k -colouring. A graph H is a *minor* of a graph G if H is isomorphic to a graph that can be obtained from a subgraph of G by contracting edges. A graph G is *H -minor-free* if H is not a minor of G . Let $\mathcal{G}_H$ denote the class of H -minor-free graphs. Hadwiger [8] famously conjectured that $\chi(G) \leq h - 1$ for every K_h -minor-free graph G . This is one of the most important open problems in graph theory. The best known upper bound on the chromatic number of K_h -minor-free graphs is $O(h \log \log h)$ due to Delcourt and Postle [1] (see [15, 21, 27] for previous bounds). It is open whether every K_h -minor-free graph is properly $O(h)$ -colourable. See [24] for a survey on this conjecture.

We consider the intersection of both a relaxation and a strengthening of Hadwiger's conjecture. First, the relaxation. For $k \in \mathbb{N}_0$, a colouring has *clustering k* if each monochromatic component has at most k vertices. The *clustered chromatic number* $\chi_\star(\mathcal{G})$ of a graph class $\mathcal{G}$ is the minimum $c \in \mathbb{N}_0$ for which there exists $k \in \mathbb{N}_0$ such that every graph $G \in \mathcal{G}$ can be c -coloured with clustering k . Similarly, the *defective chromatic number* $\chi_\Delta(\mathcal{G})$ of a graph class $\mathcal{G}$ is the minimum $c \in \mathbb{N}_0$ for which there exists $d \in \mathbb{N}_0$ such that every graph $G \in \mathcal{G}$ can be c -coloured with each monochromatic component having maximum degree at most d . See [30] for a survey on clustered and defective colouring.

Motivated by Hadwiger's conjecture, Edwards, Kang, Kim, Oum, and Seymour [6] proved a defective variant: $\chi_\Delta(\mathcal{G}_{K_h}) = h - 1$ for all $h \in \mathbb{N}$. They also introduced the *Clustered Hadwiger Conjecture*: for all $h \in \mathbb{N}$,

$$\chi_\star(\mathcal{G}_{K_h}) = h - 1.$$

Kawarabayashi and Mohar [14] had previously proved the first $O(h)$ upper bound on $\chi_\star(\mathcal{G}_{K_h})$. The constant in the $O(h)$ term was successively improved [6, 14, 17–20, 28, 29], before a proof of the Clustered Hadwiger Conjecture was first announced to appear in a future paper by Dvořák and Norin [5]. Their full proof has not yet been made public. Dujmović, Esperet, Morin, and Wood [4] recently proved the conjecture.

Now for the strengthening of Hadwiger's conjecture. Let G and H be graphs. An *H -model* in G is a collection $\mathcal{H} = (T_x : x \in V(H))$ of pairwise vertex-disjoint subtrees of G such that for each $xy \in E(H)$, there is an edge of G joining T_x and T_y . We denote by $G\langle \mathcal{H} \rangle$ the subgraph of G that is the union $\bigcup (T_x : x \in V(H))$ together with the edges in G joining T_x and T_y for each $xy \in E(H)$. Clearly H is a minor of G

if and only if there exists an H -model in G . We call $\mathcal{H}$ *odd* if there is a 2-colouring c of $G\langle\mathcal{H}\rangle$ such that:

- for each $x \in V(H)$, T_x is properly 2-coloured by c ; and
- for each $xy \in E(H)$, there is a monochromatic edge in $G\langle\mathcal{H}\rangle$ joining T_x and T_y .

We say c *witnesses* that $\mathcal{H}$ is odd. If there is an odd H -model in G , then we say that H is an *odd-minor* of G . A graph G is *H -odd-minor-free* if H is not an odd-minor of G . Let $\mathcal{G}_H^{\text{odd}}$ denote the class of H -odd-minor-free graphs.

As a qualitative strengthening of Hadwiger's conjecture, Gerards and Seymour [10, pp. 115] introduced the *Odd Hadwiger Conjecture*: $\chi(G) \leq h - 1$ for every K_h -odd-minor-free graph G . Note that in the case when $h = 3$, excluding K_3 as an odd-minor is equivalent to excluding an odd cycle as a subgraph, so the class of K_3 -odd-minor-free graphs is exactly the class of all bipartite graphs. Recently, Steiner [25] showed that the Odd Hadwiger Conjecture is asymptotically equivalent to Hadwiger's conjecture, in the sense that if every K_t -minor-free graph is properly $f(t)$ -colourable, then every K_t -odd-minor-free graph is properly $2f(t)$ -colourable.

We are interested in understanding the clustered chromatic number of graph classes defined by a forbidden odd-minor H . How does the structure of H relate to the clustered chromatic number of $\mathcal{G}_H^{\text{odd}}$? Kawarabayashi [13] first proved $\chi_*(\mathcal{G}_{K_h}^{\text{odd}}) \in O(h)$. The constant factor term in the $O(h)$ was improved in [12, 18], culminating in a result of Steiner [26] who showed that $\chi_*(\mathcal{G}_{K_h}^{\text{odd}}) \leq 2h - 2$ using chordal partitions. So $\chi_*(\mathcal{G}_H^{\text{odd}}) \leq 2|V(H)| - 2$ for every graph H . We show that the clustered chromatic number of $\mathcal{G}_H^{\text{odd}}$ is tied to the connected tree-depth of H , which we now define.

The *vertex-height* of a rooted tree T is the maximum number of vertices on a path from a root to a leaf in T . For two vertices u, v in a rooted tree T , we say that u is a *descendant* of v in T if v lies on the path from the root to u in T . The *closure* of T is the graph with vertex set $V(T)$ where $uv \in E(T)$ if u is a descendant of v or vice versa. The *connected tree-depth* $\overline{\text{td}}(G)$ of a graph G is the minimum vertex-height of a rooted tree T with $V(T) = V(G)$ such that G is a subgraph of the closure of T^* .

Ossona de Mendez, Oum, and Wood [23] showed that $\overline{\text{td}}(H)$ is a lower bound for the defective chromatic number of $\mathcal{G}_H$. In particular, for every graph H ,

$$\overline{\text{td}}(H) - 1 \leq \chi_\Delta(\mathcal{G}_H).$$

Ossona de Mendez et al. [23] conjectured that this inequality holds with equality. A partial answer to this conjecture was given by Norin, Scott, Seymour, and Wood [22], who showed that the defective chromatic number of $\mathcal{G}_H$, the clustered chromatic number of $\mathcal{G}_H$, and the connected tree-depth of H are tied. In particular,

$$\overline{\text{td}}(H) - 1 \leq \chi_\Delta(\mathcal{G}_H) \leq \chi_*(\mathcal{G}_H) \leq 2^{\overline{\text{td}}(H)+1} - 4. \quad (1)$$

* A forest is *rooted* if each component has a root vertex (which defines the ancestor relation). The *closure* of a rooted forest F is the disjoint union of the closure of each rooted component of F . The *tree-depth* $\text{td}(G)$ of a graph G is the minimum vertex-height of a rooted forest F with $V(F) = V(G)$ such that G is a subgraph of the closure of F . If G is connected, then $\text{td}(G) = \overline{\text{td}}(G)$. In fact, $\text{td}(G) = \overline{\text{td}}(G)$ unless G has two connected components G_1 and G_2 with $\text{td}(G_1) = \text{td}(G_2) = \text{td}(G)$, in which case $\overline{\text{td}}(G) = \text{td}(G) + 1$.

Norin et al. [22] also gave examples of graphs H such that $\chi_*(\mathcal{G}_H) \geq 2\overline{\text{td}}(H) - 2$, and conjectured that $\chi_*(\mathcal{G}_H) \leq 2\overline{\text{td}}(H) - 2$ for every graph H . Recently, Liu [16] proved the above conjecture of Ossona de Mendez et al. [23] which, together with a result in Liu and Oum [17], implies that for every graph H ,

$$\overline{\text{td}}(H) - 1 = \chi_\Delta(\mathcal{G}_H) \leq \chi_*(\mathcal{G}_H) \leq 3\chi_\Delta(\mathcal{G}_H) \leq 3\overline{\text{td}}(H) - 3.$$

Now consider the defective and clustered chromatic numbers of $\mathcal{G}_H^{\text{odd}}$. We prove the following analogue of the above result of Norin et al. [22] for odd-minors.

Theorem 1. *For every graph H ,*

$$\chi_*(\mathcal{G}_H^{\text{odd}}) \leq 3 \cdot 2^{\overline{\text{td}}(H)} - 4.$$

This implies that $\chi_\Delta(\mathcal{G}_H^{\text{odd}})$ and $\chi_*(\mathcal{G}_H^{\text{odd}})$ are tied to the connected tree-depth of H . In particular,

$$\begin{aligned} \overline{\text{td}}(H) - 1 = \chi_\Delta(\mathcal{G}_H) &\leq \chi_\Delta(\mathcal{G}_H^{\text{odd}}) \leq \chi_*(\mathcal{G}_H^{\text{odd}}) \leq 3 \cdot 2^{\overline{\text{td}}(H)} - 4 \quad \text{and} \\ \overline{\text{td}}(H) - 1 = \chi_\Delta(\mathcal{G}_H) &\leq \chi_*(\mathcal{G}_H) \leq \chi_*(\mathcal{G}_H^{\text{odd}}) \leq 3 \cdot 2^{\overline{\text{td}}(H)} - 4. \end{aligned}$$

We conjecture the following improved upper bounds.

Conjecture 1. *For every graph H ,*

$$\chi_*(\mathcal{G}_H^{\text{odd}}) \leq 2\overline{\text{td}}(H) - 2 \quad \text{and} \quad \chi_\Delta(\mathcal{G}_H^{\text{odd}}) = \overline{\text{td}}(H) - 1.$$

Liu and Wood [18] proved that for every graph H and for all integers $s, t \in \mathbb{N}$, every H -odd-minor-free $K_{s,t}$ -subgraph-free graph is $(2s+1)$ -colourable with clustering at most some function $f(H, s, t)$. With $s = 1$ this says that for every graph H and integer t , every H -odd-minor-free graph with maximum degree less than t is 3-colourable with clustering at most some function $f(H, t)$. This implies that $\chi_*(\mathcal{G}_H^{\text{odd}}) \leq 3\chi_\Delta(\mathcal{G}_H^{\text{odd}})$.

2 Proof

We now work towards proving [Theorem 1](#). Our proof uses the same strategy that Norin et al. [22] used to prove (1) with some modifications in the odd-minor-free setting.

We need the following definition. A *tree-decomposition* of a graph G is a collection $\mathcal{W} = (W_x : x \in V(T))$ of subsets of $V(G)$ indexed by the nodes of a tree T such that (a) for every edge $vw \in E(G)$, there exists a node $x \in V(T)$ with $v, w \in W_x$; and (b) for every vertex $v \in V(G)$, the set $\{x \in V(T) : v \in W_x\}$ induces a non-empty (connected) subtree of T . The *width* of $\mathcal{W}$ is $\max\{|W_x| : x \in V(T)\} - 1$. The *treewidth* $\text{tw}(G)$ of a graph G is the minimum width of a tree-decomposition of G . Treewidth is the

standard measure of how similar a graph is to a tree. Indeed, a connected graph has treewidth at most 1 if and only if it is a tree.

The following result due to Demaine, Hajiaghayi, and Kawarabayashi [2] allows us to work in the bounded treewidth setting.

Lemma 1 ([2]). *For every graph H , there exists a constant $w \in \mathbb{N}$ such that every H -odd-minor-free graph G has a 2-colouring where each monochromatic component has treewidth at most w .*

The following Erdős-Pósa type result is folklore (see [9, Lemma 9] which implies this).

Lemma 2. *For every graph G , for every collection $\mathcal{F}$ of connected subgraphs of G , and for every $\ell \in \mathbb{N}$, either:*

- (a) *there are ℓ vertex-disjoint subgraphs in $\mathcal{F}$, or*
- (b) *there is a set $S \subseteq V(G)$ of size at most $(\ell - 1)(\text{tw}(G) + 1)$ such that $S \cap V(F) \neq \emptyset$ for all $F \in \mathcal{F}$.*

For $h, d \in \mathbb{N}$, let $U_{h,d}$ be the closure of the complete d -ary tree of vertex-height h . Observe that $U_{h,d}$ has connected tree-depth h . Moreover, for every graph H of connected tree-depth at most h , H is isomorphic to a subgraph of $U_{h,d}$ for $d = |V(H)|$. So to prove Theorem 1, it suffices to do so for $U_{h,d}$ for all $h, d \in \mathbb{N}$.

The next definition is critical in allowing us to lift the proof of Norin et al. [22] from the minor-free setting to the odd-minor-free setting. Say an H -model $(T_x : x \in V(H))$ in G is *non-trivial* if $|V(T_x)| \geq 2$ for each $x \in V(H)$. Non-trivial models have been previously studied, for example, in [7]. The next lemma is the heart of the paper.

Lemma 3. *For all $d, h, w \in \mathbb{N}$, every graph G with treewidth at most w that does not contain a non-trivial odd $U_{h,d}$ -model is $(3 \cdot 2^{h-1} - 2)$ -colourable with clustering at most $dw + d - w$.*

Proof. We proceed by induction on h . In the base case $h = 1$, we have $U_{1,d} \cong K_1$, implying G contains no tree on at least two vertices, and thus $E(G) = \emptyset$. Hence G can be coloured with $1 = 3 \cdot 2^{1-1} - 2$ colour with clustering 1.

Now assume $h \geq 2$ and the claim holds for $h - 1$. Without loss of generality, we may assume that G is connected. Fix $r \in V(G)$. For each $i \in \mathbb{N}_0$, let $V_i := \{v \in V(G) : \text{dist}_G(v, r) = i\}$. We refer to each V_i as a *layer*. Fix $i \geq 1$ where $V_i \neq \emptyset$, and let $u_i \in V_i$. Let $\mathcal{M}$ be the collection of all non-trivial odd $U_{h-1,d}$ -models in $G[V_i \setminus \{u_i\}]$, and let $\mathcal{F} := \{G\langle \mathcal{H} \rangle : \mathcal{H} \in \mathcal{M}\}$. Each element of $\mathcal{F}$ is a connected subgraph of $G[V_i \setminus \{u_i\}]$ since $U_{h-1,d}$ is connected.

Suppose that $\mathcal{F}$ contains d pairwise vertex-disjoint subgraphs $G\langle \mathcal{H}_1 \rangle, \dots, G\langle \mathcal{H}_d \rangle$. Then each $\mathcal{H}_j$ is a non-trivial odd $U_{h-1,d}$ -model in $G[V_i \setminus \{u_i\}]$. Say $\mathcal{H}_j = (T_x^{(j)} : x \in V(U_{h-1,d}))$. Let c_j be a red-blue colouring of $G\langle \mathcal{H}_j \rangle$ that witnesses $\mathcal{H}_j$ being odd. Since $\mathcal{H}_j$ is non-trivial, $T_x^{(j)}$ contains both a red and a blue vertex for each $x \in V(U_{h-1,d})$. Let T be a spanning tree of $G[V_0 \cup \dots \cup V_{i-1} \cup \{u_i\}]$ rooted at r , where for each non-root vertex $u \in V(T) \cap V_j$ there is exactly one edge uw in T with $w \in V(T) \cap V_{j-1}$. Then $|V(T)| \geq 2$. Moreover, there is a proper 2-colouring

c_T of T where each vertex in V_{i-1} is coloured red. Since every vertex in V_i has a neighbour in V_{i-1} , for each $x \in V(H)$ and $j \in [d]$, each red vertex in $T_x^{(j)}$ is adjacent to a red vertex in T . Observe that $U_{h,d}$ can be constructed by taking d disjoint copies of $U_{h-1,d}$ plus a dominant vertex. By taking $\mathcal{H}_1, \dots, \mathcal{H}_d$ together with T and all the edges between $G\langle\mathcal{H}_1\rangle \cup \dots \cup G\langle\mathcal{H}_d\rangle$ and V_{i-1} , it follows that G contains a non-trivial $U_{h,d}$ -model $\mathcal{H} = (T_x: x \in V(U_{h,d}))$. Furthermore, let $c(v) := c_T(v)$ for all $v \in V(T)$ and $c(w) := c_j(w)$ for all $w \in V(G\langle\mathcal{H}_j\rangle)$ and $j \in [d]$. Then c witnesses that $\mathcal{H}$ is odd, a contradiction.

Thus $\mathcal{F}$ does not contain d pairwise vertex-disjoint subgraphs. By [Lemma 2](#) applied to $\mathcal{F}$, there is a set $S_i \subseteq V_i$ such that $|S_i| \leq (d-1)(w+1)$ and $G[V_i \setminus (S_i \cup \{u_i\})]$ contains no subgraph in $\mathcal{F}$. Hence $G[V_i \setminus (S_i \cup \{u_i\})]$ contains no non-trivial odd $U_{h-1,d}$ -model. By induction, $G[V_i \setminus (S_i \cup \{u_i\})]$ can be $(3 \cdot 2^{h-2} - 2)$ -coloured with clustering $dw + d - w$. Use a new colour for the vertices in $S_i \cup \{u_i\}$. So $G[V_i]$ can be $(3 \cdot 2^{h-2} - 1)$ -coloured with clustering at most $dw + d - w$. Moreover, $G[V_0]$ can be 1-coloured with clustering 1 since $|V_0| = 1$. By alternating the colour sets used for each layer, it follows that G can be coloured with $2(3 \cdot 2^{h-2} - 1) = 3 \cdot 2^{h-1} - 2$ colours and with clustering at most $dw + d - w$, as required. $\square$

We are now ready to prove our main result.

Proof (of [Theorem 1](#)). Let G be an H -odd-minor-free graph. Then G is $U_{\overline{\text{td}}(H),d}$ -odd-minor-free for $d = |V(H)|$. By [Lemma 1](#), there exists an integer $w = w(\overline{\text{td}}(H), d)$ such that G has a red–blue colouring where each monochromatic component of G has treewidth at most w . Since G is $U_{\overline{\text{td}}(H),d}$ -odd-minor-free, G contains no non-trivial odd $U_{\overline{\text{td}}(H),d}$ -model. By [Lemma 3](#), each monochromatic component of G can be $(3 \cdot 2^{\overline{\text{td}}(H)-1} - 2)$ -coloured with clustering $dw + d - w$. Using distinct colour sets for the red and blue components of G , we conclude that G can be $(3 \cdot 2^{\overline{\text{td}}(H)} - 4)$ -coloured with clustering $dw + d - w$. Hence, $\chi_\star(\mathcal{G}_H^{\text{odd}}) \leq 3 \cdot 2^{\overline{\text{td}}(H)} - 4$. $\square$

Acknowledgements [Theorem 1](#) was first published in the Ph.D. thesis of the second author [11]. This result was independently discovered at the IBS-MATRIX “[Structural Graph Theory Downunder III](#)” workshop at the Mathematical Research Institute MATRIX (April 2023).

References

- [1] MICHELLE DELCOURT AND LUKE POSTLE. [Reducing linear Hadwiger’s conjecture to coloring small graphs](#). 2021, arXiv:2108.01633.
- [2] ERIK D. DEMAINE, MOHAMMADTAGHI HAJIAGHAYI, AND KEN-ICHI KAWARABAYASHI. [Decomposition, approximation, and coloring of odd-minor-free graphs](#). In *Proc. 21st Annual ACM-SIAM Symp. on Discrete Algorithms (SODA ’10)*, pp. 329–344. ACM Press, 2010.
- [3] REINHARD DIESTEL. [Graph theory](#), vol. 173 of *Graduate Texts in Mathematics*. Springer, 5th edn., 2017.

- [4] VIDA DUJMOVIĆ, LOUIS ESPERET, PAT MORIN, AND DAVID R. WOOD. [Proof of the clustered Hadwiger conjecture](#). In *Proc. 64th IEEE Annual Symposium on Foundations of Computer Science (FOCS '23)*, pp. 1921–1930. 2023. arXiv:2306.06224.
- [5] ZDENĚK DVOŘÁK AND SERGEY NORIN. [Islands in minor-closed classes. I. Bounded treewidth and separators](#). 2017, arXiv:1710.02727.
- [6] KATHERINE EDWARDS, DONG YEAP KANG, JAEHOON KIM, SANG-IL OUM, AND PAUL SEYMOUR. [A relative of Hadwiger's conjecture](#). *SIAM J. Discrete Math.*, 29(4):2385–2388, 2015.
- [7] SAMUEL FIORINI, GWENAËL JORET, DIRK OLIVER THEIS, AND DAVID R. WOOD. [Small minors in dense graphs](#). *European J. Combin.*, 33(6):1226–1245, 2012.
- [8] HUGO HADWIGER. Über eine Klassifikation der Streckenkomplexe. *Vierteljschr. Naturforsch. Ges. Zürich*, 88:133–142, 1943.
- [9] FREDDIE ILLINGWORTH, ALEX SCOTT, AND DAVID R. WOOD. [Product structure of graphs with an excluded minor](#). 2022, arXiv:2104.06627. *Trans. Amer. Math. Soc.*, to appear.
- [10] TOMMY R. JENSEN AND BJARNE TOFT. Graph coloring problems. John Wiley, 1995.
- [11] DONG YEAP KANG. [Graph decompositions and related extremal problems](#). Ph.D. thesis, KAIST, 2020.
- [12] DONG YEAP KANG AND SANG-IL OUM. [Improper coloring of graphs with no odd clique minor](#). *Combin. Probab. Comput.*, 28(5):740–754, 2019.
- [13] KEN-ICHI KAWARABAYASHI. [A weakening of the odd Hadwiger's conjecture](#). *Combin. Probab. Comput.*, 17(6):815–821, 2008.
- [14] KEN-ICHI KAWARABAYASHI AND BOJAN MOHAR. [A relaxed Hadwiger's conjecture for list colorings](#). *J. Combin. Theory Ser. B*, 97(4):647–651, 2007.
- [15] ALEXANDR V. KOSTOCHKA. [Lower bound of the Hadwiger number of graphs by their average degree](#). *Combinatorica*, 4(4):307–316, 1984.
- [16] CHUN-HUNG LIU. [Defective coloring is perfect for minors](#). *Combinatorica*, 44(3):467–507, 2024.
- [17] CHUN-HUNG LIU AND SANG-IL OUM. [Partitioning \$H\$ -minor free graphs into three subgraphs with no large components](#). *J. Combin. Theory Ser. B*, 128:114–133, 2018.
- [18] CHUN-HUNG LIU AND DAVID R. WOOD. [Clustered coloring of graphs excluding a subgraph and a minor](#). 2019, arXiv:1905.09495.
- [19] CHUN-HUNG LIU AND DAVID R. WOOD. [Clustered variants of Hajós' conjecture](#). *J. Combin. Theory Ser. B*, 152:27–54, 2022.
- [20] SERGEY NORIN. Conquering graphs of bounded treewidth. 2015. Unpublished manuscript.
- [21] SERGEY NORIN, LUKE POSTLE, AND ZI-XIA SONG. [Breaking the degeneracy barrier for coloring graphs with no \$K_t\$ minor](#). *Adv. Math.*, 422:109020, 2023.
- [22] SERGEY NORIN, ALEX SCOTT, PAUL SEYMOUR, AND DAVID R. WOOD. [Clustered colouring in minor-closed classes](#). *Combinatorica*, 39(6):1387–1412, 2019.

- [23] PATRICE OSSONA DE MENDEZ, SANG-IL OUM, AND DAVID R. WOOD. [Defective colouring of graphs excluding a subgraph or minor](#). *Combinatorica*, 39(2):377–410, 2019.
- [24] PAUL SEYMOUR. [Hadwiger’s conjecture](#). In JOHN FORBES NASH JR. AND MICHAEL TH. RASSIAS, eds., *Open Problems in Mathematics*, pp. 417–437. Springer, 2016.
- [25] RAPHAEL STEINER. [Asymptotic equivalence of Hadwiger’s conjecture and its odd minor-variant](#). *J. Combin. Theory Ser. B*, 155:45–51, 2022.
- [26] RAPHAEL STEINER. [Improved bound for improper colourings of graphs with no odd clique minor](#). *Combinatorics, Probability and Computing*, 32(2):326–333, 2023.
- [27] ANDREW THOMASON. [An extremal function for contractions of graphs](#). *Math. Proc. Cambridge Philos. Soc.*, 95(2):261–265, 1984.
- [28] JAN VAN DEN HEUVEL AND DAVID R. WOOD. [Improper colourings inspired by Hadwiger’s conjecture](#). *J. London Math. Soc.*, 98:129–148, 2018. [arXiv:1704.06536](#).
- [29] DAVID R. WOOD. [Contractibility and the Hadwiger conjecture](#). *European J. Combin.*, 31(8):2102–2109, 2010.
- [30] DAVID R. WOOD. [Defective and clustered graph colouring](#). *Electron. J. Combin.*, DS23, 2018.